

Machine Learning For Design Engineers

Week 1: Introduction

Pietro Ferraro

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- 4 What is Deep Learning
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This course is divided into two parts. The first one will go over the basic theoretical ideas of Deep Learning, the second one will go over the main applications.

Who we are

Module Leaders



- I am a lecturer in Artificial Intelligence at the Dyson School of Design Engineering and at I-X. My research interests include control theory applied to the sharing economy domain and Reinforcement Learning (RL).
- Email: **p.ferraro@imperial.ac.uk**

Who we are

Module Leaders

- Professor Robert Shorten, Co-leader
- Amelia Bryant, GTA
- Haozhe Tian, GTA
- Abigail Langbridge, GTA
- Micheal Vitali, GTA
- Felix wieberneit, GTA

Logistics of the course

- The course will be delivered on **Tuesdays** from **16:00** to **18:00**
- On **Thursdays**, from **11:00** to **13:00** you have a online tutorial where you will be divided into groups and will have the chance to ask the GTAs and the module leaders about the content of the class and about the labs.
- Drafts of Lectures and Labs will be available on Blackboard in advance.

What is Machine Learning?

What is Machine Learning?

According to the IBM website:

‘Machine learning (ML) is a branch of artificial intelligence (AI) and computer science that focuses on the using data to enable algorithms to imitate the way that humans learn, gradually improving its accuracy.’

According to the Oxford dictionary:

‘The use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data.’

What is Machine Learning?

If these definitions appear broad and vague, it is because they are.

What is Machine Learning?

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Topics that fall under this definition:

- Regression
- Clustering
- Statistical inference
- Deep Learning
- Reinforcement Learning
- Data Science

What is Machine Learning?

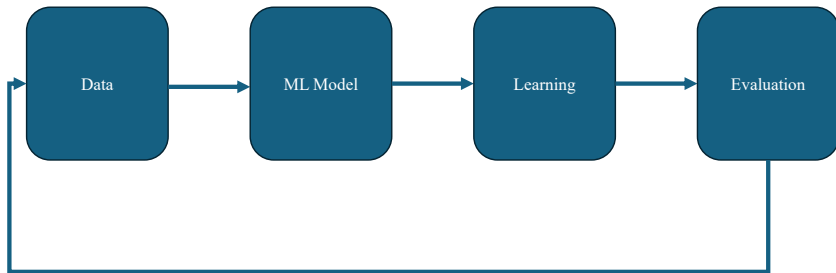
In general one can think of a ML model as something like this (broadly)

- A set of data, called Training Data, is used to train a model.
- The model learns patterns in the data.
- The performance of the model is evaluated and, based on this performance, the cycle repeats

This will keep going until a desired level of performance is reached (if possible).

What is Machine Learning?

In general one can think of a ML model as something like this (broadly)



What is Machine Learning?

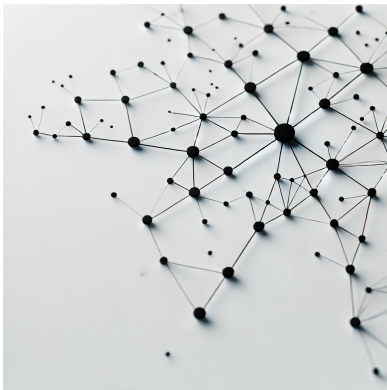
This course will focus on several areas of ML, called Deep Learning (DL).

- Regression
- Clustering
- Statistical inference
- **Deep Learning**
- **Reinforcement Learning**
- Data Science

What is Deep Learning?

What is Deep Learning?

A brief history - Invention of Neural Networks



Deep learning (DL) is a subset of machine learning that uses multilayered neural networks (NN), called deep neural networks, to simulate the complex decision-making power of the human brain.

What is Deep Learning?

A brief history - Invention of Neural Networks

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH and WALTER H. PITTS

Because of the “all-or-none” character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

The emergence of DL as a field starts approximately in 1943, when Warren McCulloch and mathematician Walter Pitts wrote a paper on how neurons might work. In order to do so, they modeled a simple neural network using electrical circuits.

What is Deep Learning?

A brief history - Pivotal moments

- **1950s** Advancement of computers
- **1959** Bernard Widrow and Marcian Hoff of Stanford developed neurons called "ADALINE". This was the first neural network (NN) application to a real world problem, whose aim was to eliminate echoes on phone lines.
- **1986** Back-Propagation is formalised by three different research groups, simultaneously.
- **Early 2000** Development of massive computational power and GPUs brings neural networks back from their slumber.
- **2010** A paper from Nair and Hinton popularizes the use of ReLu as an activation function.
- **2012** Paper on ImageNet popularizes Convolutional Neural Networks (CNNs).

What is Deep Learning?

Problems where DL makes a difference

Deep Learning is great at:

- Prediction problems;
- Pattern recognition problems;
- Problems that involve sequential decision making;
- Generating data with similar properties to the training dataset

What is Deep Learning?

Prediction problems



Making predictions about how a certain variable will evolve with time

Examples:

- Predicting weather
- Predicting stock market

What is Deep Learning?

Pattern recognition



Identifying objects in images

Examples:

- Identify plug-in vehicles from images
- Segment images
- Distinguish between electric vehicles and internal combustion ones

What is Deep Learning?

Sequential decision making



Learn a sequence of actions to achieve a desired result in an environment.

Examples:

- Play chess or Go or the Atari games
- Learn how to synthesize proteins
- Control the magnetic field of a nuclear fusion reactor

What is Deep Learning?

Generative AI



Generating data with similar properties to the training dataset.

Examples:

- LLMs, like chatGPT or Gemini
- Image generation systems, like Midjourney

How will the course be delivered?

How will the course be delivered?

List of topics

The course is divided into two sections. The first one will cover the following topics:

- Neurons and their properties
- Neural Networks as universal approximators
- Training and back-propagation
- Activation functions and loss functions
- Convolutional Neural Networks and image recognition

How will the course be delivered?

List of topics

The second part will delve into more advanced topics and will be taught by guest lecturers:

- Markov Chains
- MDPs and Reinforcement Learning
- Transformers and LLMs
- Optimal Transport or maybe another topic (GNNs? Review?)

How will the course be delivered?

Prerequisites

While the course has no formal prerequisites there are a number of topics that you will need to be familiar with, in order to engage successfully with the material:

- Basics of calculus (in particular, derivatives and the chain rule).
- Basics of linear algebra (in particular operations with matrices and vectors)
- Programming (knowing how to use variables, for loops, if/then/else, etc., knowledge of OOP is not required)

We will not go over these topics in the course but we will provide plenty of materials and references on Blackboard for you to consult and study.

How will the course be delivered?

Lectures and Laboratories

Lectures will be delivered in person on Tuesdays and from week 3 we will start the laboratories:

- The laboratories are designed to make you apply the topics that you learnt in class.
- Each lab is self-contained and you will complete it at home.

How will the course be delivered?

Lectures and Laboratories

- Software: Python 3.11 and above.
- You can use any Python distribution you prefer. The one we recommend is visual studio code <https://code.visualstudio.com>
- Required libraries: Numpy, Torch, Torchvision. For Torch and Torchvision, please use the following link <https://pytorch.org/> (select Conda or Pip, depending on which one you are using and for "compute platform" select CPU.)
- **Please, make sure that all the required software is installed before Labs start**

How will the course be delivered?

The Clinics

Tutorials will be held online. The way they will work is that you will receive the lab instructions and you will be assigned to a Teams room with a TA. During the two hours you will be able to ask the TA questions about the lab instructions and/or clarifications about the theory covered in class up to that point.

How will the course be delivered?

Exams and Assignments

Assessment will have three components.

- An **end of term exam** which will constitute 70% of the final grade. This will be a 30 minutes viva.
- A **short oral examination** based on the completed labs. This will constitute 15% of the final grade. This will be a 5-10 minutes viva.
- **Oral Presentation**. This will constitute 15% of the final grade. This will be a 10 minutes viva.

References

How will the course be delivered?

1. The hundred-page machine learning book, A. Burkov
2. Hands-on machine learning with SciKit-learn, Keras and TensorFlow, A. Geron
3. Decision making under uncertainty, M. Koyhendorfer
4. Google's page rank and beyond, A. Langville, C. Meyer
5. Mathematics for Machine Learning, M. Diefenroth et. al.

How will the course be delivered?

Studying material and feedback

All the material will be provided on Blackboard.

- You will have access to the lecture of the week and the lab, on Monday of the same week, through Blackboard (although notice that until the day of the lecture there might be changes to the slides).
- Each week you will be provided with supplementary material in order to further explore the topics covered during lectures.

We will make use of Menti, to regularly collect feedback on the course and to adjust our teaching style.

How will the course be delivered?

Contacts

You can always contact us, on Teams and via email. Please, be mindful that it might take more than one day to reply to you (especially if you contact us outside of working hours and in the weekend!).

- p.ferraro@imperial.ac.uk
- r.shorten@imperial.ac.uk

What is Machine Learning?

What is machine learning?

Machine learning is about model building. Models are just compact mathematical representations of data. All models are wrong, but some are useful (George Box).

In machine learning we build a model to solve a problem:

- by gathering some real data;
- and then build a mathematical model of some phenomena based on the gathered data.

ML algorithms are often classified according to the manner in which the model is built (supervised learning, unsupervised learning, reinforcement learning, graph based algorithms)

Types of machine learning

Supervised learning

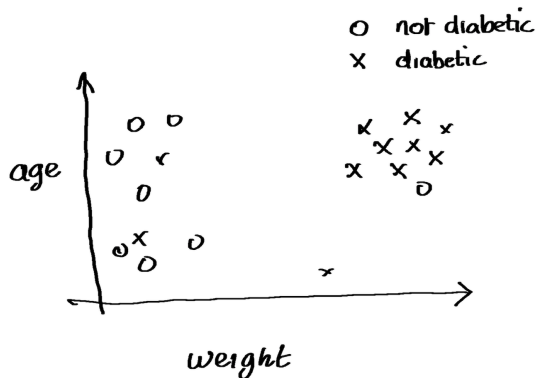
In supervised learning, the dataset is a collection of labelled examples $(x(i), y(i))$, $i \in 1, \dots, N$. Each element $x(i)$ is called a feature vector, and each $y(i)$ is a label. Typically, the $x(i)$, $y(i)$ could be vectors.

The goal of a supervised learning algorithm is to produce an algorithm such that given a feature vector x , we can produce a label y . Typically, we have two types of problems in supervised learning:

- Classification
- Regression

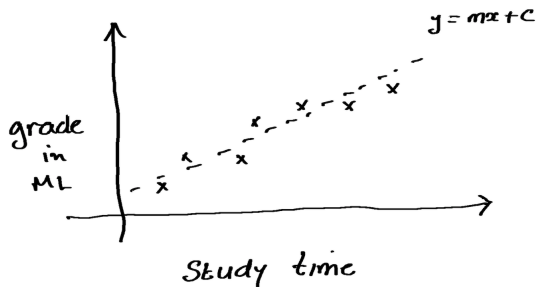
Supervised learning: classification

For example, suppose that $x(i)$ was a vector that recorded age and weight of an individual, and $y(i)$ recorded whether this individual was diabetic. Could we predict diabetes given an $x(i)$?



Supervised learning: regression

Suppose now that $x(i)$ was a vector that recorded time an individual studied, and $y(i)$ recorded the final grade in ML. Could we predict success in ML class given an $x(i)$?



Un-supervised learning

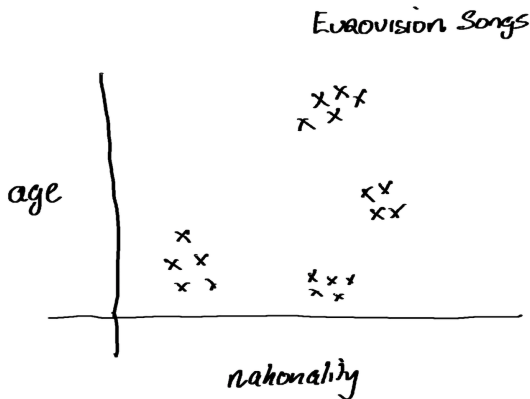
In unsupervised learning the training data is not labeled. Here the system tries to learn without a (explicit) teacher.

Adding labels to data is very expensive. Usually humans have to do it. Think of taking pictures to monitor a machine.

Unsupervised data-sets are by far less expensive than supervised ones.

Un-supervised learning

Used in clustering applications; anomaly detection; density estimation applications.



Semi-supervised learning

In semi-supervised learning, the dataset contains both labeled and un-labeled examples. The goal of the semi-supervised learning is the same as in supervised learning.

An example of un-supervised learning is services such as Google photo.

Images are clustered according to faces appearing in the image. This is the unsupervised part of the algorithm.

Some images are labeled - this is Bob Shorten. Any cluster with Bob Shorten then has an identifier.

Reinforcement learning

A completely different ML system is called reinforcement learning.

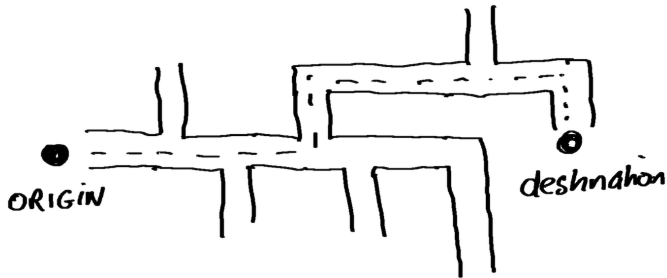
Reinforcement learning is really useful in Robotic applications, or in traffic applications.

Here the learning system actively probes the environment, and is rewarded, or punished, based on the outcome.

The objective is then to find strategy, or policy, that maximizes the reward.

Reinforcement learning

Imagine an autonomous vehicle, trying to find the best route to a destination. The route with the shortest path maximizes the reward.



Reinforcement learning

Reinforcement learning has been amazingly successful in many applications.

- Robotics
- Games - Chess, Go.
- Control of complex dynamical systems.

Mathematics of Machine Learning

Mathematical background

So what's needed to do some ML? ML is built on four areas of mathematics.

- 1 Optimization
- 2 Probability
- 3 Linear algebra
- 4 Calculus (derivatives, chain rule)
- 5 Markov chains

We will spend a little bit of time going over [1]. For [2], [3] and [4] we will provide some extra reading material to revise the basic concepts. [5] is required for Reinforcement Learning, which we will not cover in this course.

Notation

First some notation. We will try to keep to the following convention.

- Matrices variables are denoted in capital letters - for example, the matrix P .
- Vectors are denoted in lower case, with a subscript denoting the element of a matrix - for example $w^T = (w_1, \dots, w_n)$.
- Sometimes vectors are indexed, using for example, time. We denote this by: $w(i)^T = (w_1(i), \dots, w_n(i))$ meaning the vector w at time index i .

In our context, one can think of subscripts denoting variables, and parentheses denoting measured data points.

Optimization

In Deep Learning we are often interested in finding the best parameters in order to achieve a certain goal. This is usually expressed as an optimization problem

$$\text{minimize} \quad f(x)$$

Where $f : \mathbb{R}^d \rightarrow \mathbb{R}$. The goal here is to find the value x^* such that

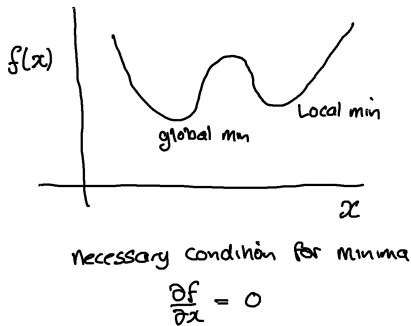
$$f(x^*) = \min\{f(x), x \in \mathbb{R}^d\}$$

The **necessary** condition for $\hat{x}$ to be a minimum is that

$$\nabla f(\hat{x}) = 0$$

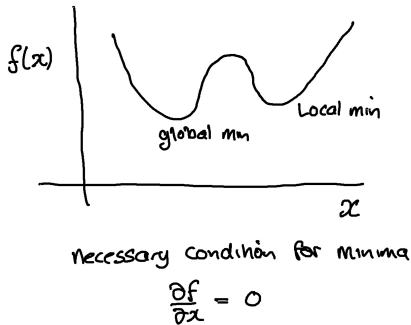
This is not sufficient to conclude that $\hat{x} = x^*$.

Optimization



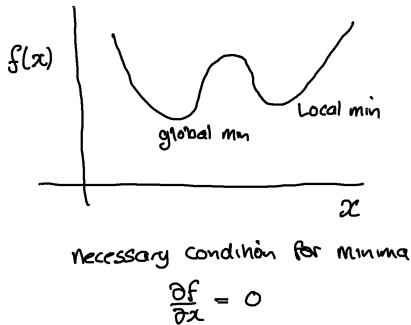
The notion of a **gradient** is central to many aspects of machine learning (necessary condition for minimum/maximum).

Gradient Descent



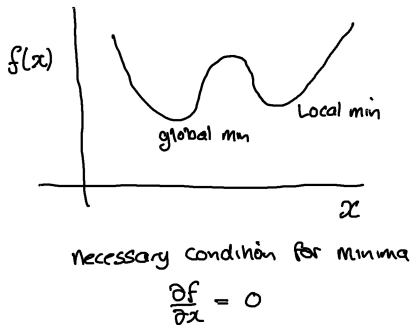
Probably the most famous numerical algorithm (and certainly the one that has the most influenced DL) is called **gradient descent**.

Gradient Descent



Imagine a ball at point x_0 . The ball will start rolling in the direction opposite to where the gradient sign is pointing at.

Gradient Descent

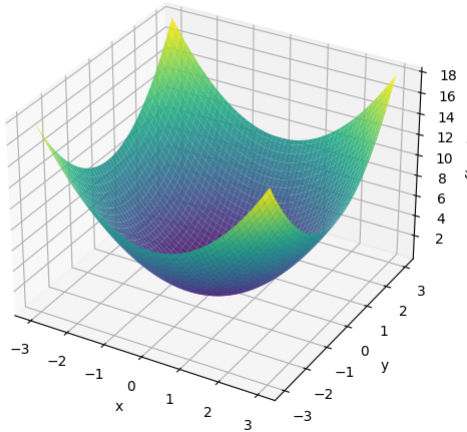


This intuition leads to the following expression (which can be proven to find a stationary point):

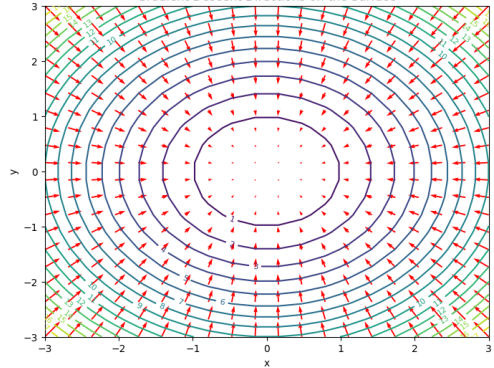
$$x_{k+1} = x_k - \eta \nabla f(x_k)$$

Gradient Descent

3D Surface Plot of $f(x, y) = x^2 + y^2$

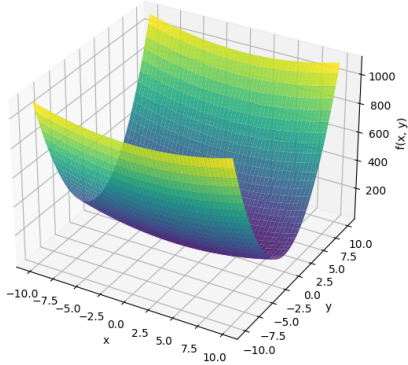


Gradient Descent Directions on the Surface

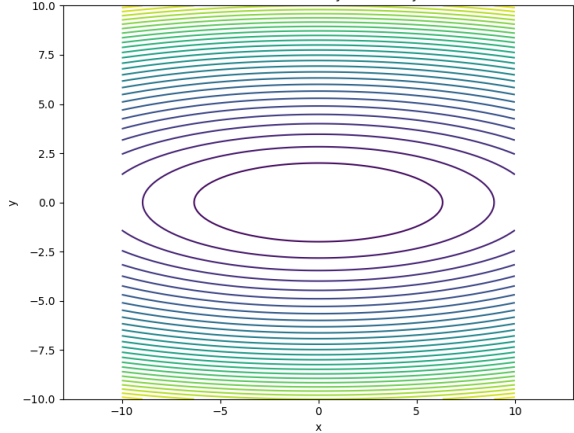


Gradient Descent - drawbacks

3D Surface Plot of $f(x, y) = x^2 + 10y^2$



Contour Plot of $f(x, y) = x^2 + 10y^2$



Gradient Descent- alternatives

There are many alternatives to gradient descent. A common one, when dealing with badly conditioned functions (almost the totality of functions in DL are badly conditioned), used to be to add an extra term, called momentum:

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathbf{x}_k - \eta \nabla \mathbf{f}(\mathbf{x}_k) + \alpha \Delta \mathbf{x}_k \\ \Delta \mathbf{x}_k &= \alpha \Delta \mathbf{x}_{k-1} - \eta \nabla \mathbf{f}(\mathbf{x}_{k-1})\end{aligned}$$

Other, more modern, approaches are:

- RMSprop
- Adam

Probability theory

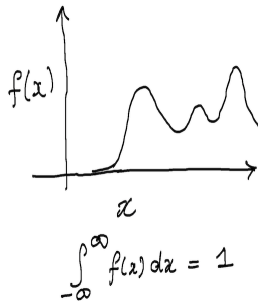
Random variable: A variable whose outcome is determined by a random event. There are two types of random variable - discrete and continuous.

A discrete random variable takes on a countable set of values such as red, blue,

The probability distribution of a discrete random variable is described by a list of probabilities, all of which sum to 1. Sometimes this function is called a **probability mass function**.

Probability theory

A continuous random variable (CRV) takes on an infinite number of values in some interval. The probability density function of a CRV is described by a probability density function.



Some basic concepts

For the discrete random variable X:

Expectation: $E(X) = \sum_{i=1}^k x(i)p(x(i))$. $E(X)$ is often denoted μ . **Variance:** $\text{var}(x) = E((X - \mu)^2)$. $\text{var}(x)$ is often denoted by σ^2 .

For the continuous random variable X:

Expectation: $E(X) = \int_{-\infty}^{\infty} xf(x)dx$. (X) is often denoted μ . **Variance:** As before $\text{var}(x) = E((X - \mu)^2)$. $\text{var}(x)$ is often denoted by σ^2 .

Jargon busting

Jargon busting

Regression: Predicting a real value given some information

Classification: Attaching labels to things.

Models: A compact description of a data set.

Parametric models: A models with class parameters - $y = mx + c$.

Non-parametric models: Models with no parameters - $y = \text{mean}(x_1 + x_2 + \dots)$.

Jargon busting

Bias: Models that favour a group (don't generalise well)

Bias - variance trade-off: Trade off between generalisation and representing training set.

Training and test set: Data used to train and test the ML tool.

Regularisation: Constraining a model with some prior information.

Over-fitting: Learning the data-set too well. (Question - can you guess why this is a problem?)

Jargon busting

Supervised learning: Using data-labels as part of learning scheme

Unsupervised learning: Learning without data-labels.

Semi-supervised learning: Some data-points have labels. Others do not.

Reinforcement learning: Learning how to make sequences of decisions.